

Robust Reinforcement Learning for Stock Trading in Non-Stationary Markets: A Case Study on the Vietnamese VN30 Index

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Abstract—Reinforcement Learning (RL) has shown strong potential for automated trading thanks to its ability to handle highly nonlinear and volatile markets. However, most existing studies focus on developed markets, leaving a notable gap in emerging markets such as Vietnam. This paper focuses on Deep SARSA for stock trading on the Vietnamese market, and compares its performance with two representative RL baselines: DQL and Policy Gradient. A simulated trading environment is built as a Markov Decision Process (MDP), where the state feature vector integrates technical indicators including MACD, RSI, CCI, and ADX. The models are tested in two contrasting scenarios: a stable bullish phase and a downturn phase. Results show that Deep SARSA, with its on-policy update mechanism, provides better risk control and stability, particularly during bearish periods. It maintains lower portfolio drawdowns and a smoother equity growth curve than DQL and PG. In favorable market conditions, all three algorithms generate profits, but Deep SARSA still exhibits superior stability relative to DQL. Overall, the study finds Deep SARSA to be a robust automated trading solution for the Vietnamese market and provides an experimental foundation for future RL research in emerging markets.

Index Terms—Reinforcement Learning, Deep SARSA, Deep Q-Learning, Policy Gradient, Automated Trading, Vietnamese Stock Market.

I. INTRODUCTION

In recent years, rapid advances in artificial intelligence (AI) have driven profound changes across many socio-economic sectors, particularly in the financial industry [1]. The growth of data availability, computational power, and learning algorithms has strongly accelerated the development of automated trading systems. Among these methods, Reinforcement Learning (RL) has emerged as a highly promising approach because it models sequential decision-making in dynamic and uncertain environments [2]. In RL, an agent learns to make optimal decisions by continuously interacting with the market environment: observing the state, taking actions such as buying, selling, or holding, and receiving rewards in the form of risk-adjusted profits or losses. Traditional approaches rely on static labeled data and fail to capture the temporal dependencies underlying financial decision-making. In contrast, RL can naturally

model sequential actions over time within highly volatile and uncertain market environments. As a result, RL has been increasingly applied in portfolio management, high-frequency trading (HFT), and risk management.

Most recent studies focus on developed markets such as the United States, Europe, and China [3]. The Vietnamese stock market is highly volatile, with unstable liquidity and frequently changing regulations. As a result, RL models trained on foreign market data often do not transfer well to Vietnam's market conditions. Prior work mainly examines value-based methods like Deep Q-Learning (DQL) [4], while on-policy models such as Deep SARSA are rarely studied in emerging markets. Because Deep SARSA updates from the agent's executed policy, it may suit non-stationary conditions typical of Vietnam.

The objective of this study is to apply and evaluate the performance of the Deep SARSA algorithm for stock trading on the Vietnamese stock market using real historical data from the VNStock library. Specifically, the study has three goals: (i) to implement and adapt a Deep SARSA model tailored to the characteristics of the Vietnamese financial market; (ii) to compare its effectiveness with baseline algorithms such as Deep Q-Learning (DQL) and Policy Gradient; and (iii) to analyze model performance across different market phases, including the "strong growth" period before 2022 and the "correction" period after 2022.

This study makes the following contributions to RL research:

- Establishing a standardized reinforcement learning (RL) environment for the Vietnamese stock market;
- Developing a comprehensive data preprocessing and feature extraction pipeline;
- Providing reproducible experiments with an open-source implementation;
- Conducting detailed quantitative analysis of profit-risk performance;
- Proposing practical directions for further RL-based trading research in Vietnam.

II. LITERATURE REVIEW

Reinforcement learning (RL) has been increasingly applied to automated trading systems because of its ability to learn optimal policies through continual interaction with the market environment [2]. In this setting, an RL agent learns buy, sell, or hold decisions from market states (e.g., prices, trading volume, technical indicators, macroeconomic variables) with the objective of maximizing long-term cumulative reward, typically defined as risk-adjusted return.

Many prior studies have shown that RL outperforms passive strategies such as buy-and-hold in nonlinear, non-stationary, and noisy market environments. Specifically, in the study by Li et al. [5], Deep Q-Learning was applied to U.S. stock trading and achieved a higher Sharpe ratio compared to the moving-average strategy. Similarly, Jiang et al. [6] developed a Deep Reinforcement Learning framework for portfolio management, demonstrating its ability to adapt to different market regimes (bull/bear/sideways) without the need for manual retraining. Recently, the emergence of specialized open-source libraries for finance has significantly enhanced the applicability and reproducibility of Reinforcement Learning. Among these, the most notable is FinRL [7], which provides a standardized Gym-based trading environment—integrated with real-world data, automated preprocessing, a suite of RL algorithms (DQL, PPO, A2C, SAC), and financial performance metrics. FinRL has been widely used in more than 100 studies (as of 2025), ranging from single-stock trading to portfolio optimization and dynamic risk management. However, most current applications still focus on developed markets, and implementation in emerging markets such as Vietnam remains limited due to the lack of realistic simulation environments and high-quality data.

Among value-based RL algorithms, Deep Q-Learning (DQL) and Deep SARSA are two representative methods, both employing deep neural networks (DNNs) to approximate the action-value function $Q(s, a)$ —a formulation introduced and advanced by Mnih et al. [4]. However, they differ fundamentally in their policy update strategies, leading to distinct behaviors and performance in real financial environments.

This difference is clearly reflected in the value-update mechanism. For DQL, the algorithm uses the assumed optimal action (off-policy) through the max function:

$$q_{\text{new}}(s_t, a_t) = q(s_t, a_t) + \alpha (R_t + \gamma \max_{a'} q(s_{t+1}, a') - q(s_t, a_t)).$$

In contrast, Deep SARSA is an on-policy algorithm, updating the value based on the actual action selected by the agent according to the current policy:

$$q_{\text{new}}(s_t, a_t) = q(s_t, a_t) + \alpha (R_t + \gamma q(s_{t+1}, a_{t+1}) - q(s_t, a_t)).$$

To balance exploitation and exploration, both algorithms use the ϵ -greedy policy, in which the action is selected

according to the probability distribution:

$$\pi(s, a) = \begin{cases} 1 - \epsilon + \frac{\epsilon}{|\mathcal{A}|}, & \text{if } a = a^*, \\ \frac{\epsilon}{|\mathcal{A}|}, & \text{otherwise,} \end{cases} \quad a^* = \arg \max_{a'} q(s, a').$$

This action-selection mechanism helps the agent avoid getting “trapped” in a local strategy and supports steady learning. Because Deep SARSA updates on-policy, it is often more stable in non-stationary settings where rewards change over time—common in real financial markets—so it suits noisy, highly volatile markets such as Vietnam. Comparative studies show clear differences: DQL on S&P 500 performs well with large, stable datasets and long-term trends [8], while Deep SARSA can outperform DQL and Policy Gradient in bearish markets [9]. Despite this, applications of Deep SARSA in emerging markets remain limited, particularly in Vietnam.

Emerging markets such as China, India, and Vietnam possess distinct characteristics compared to developed markets, including high price volatility, unstable liquidity, and frequently changing regulatory policies (e.g., capital controls in China, regulatory reforms in India, or price-band adjustments in Vietnam). These factors give rise to highly nonlinear market dynamics, heavily shaped by investor sentiment and localized macroeconomic shocks [10]. These conditions often cause traditional machine learning models to overfit or struggle with adaptability. Several prior studies have applied reinforcement learning in this context, but most focused on Deep Q-Learning or Policy Gradient rather than on-policy methods such as Deep SARSA. Specifically, Zhang et al. [11] applied Deep Q-Learning models to trade 50 futures contracts and showed that DQL outperformed traditional strategies such as momentum and MACD in terms of risk-adjusted returns during the 2011–2019 period. By using volatility-scaling mechanism, DQL maintained stable performance even under high transaction costs. In Vietnam, the Proximal Policy Optimization (PPO) was implemented to forecast and trade financial bubbles on the VN-Index and used machine learning to detect bubble periods from 2001–2021, showing that historical data remains limited for training RL models [12].

However, the above studies mainly explore Deep Q-Learning or Policy Gradient, whereas Deep SARSA, which uses an on-policy update mechanism based on actual actions, remains underexplored. This creates a clear research gap. Therefore, this study aims to address that gap by implementing, training, and evaluating the Deep SARSA model on real VNStock data, while also comparing it with classical RL methods (DQL and PG) to clarify the model’s performance, adaptability, and stability across different market phases.

III. METHODOLOGY

A. Experimental Environment

The experiments are implemented in Python and are based on the Markov Decision Process (MDP) framework, in which

each state s_t , action a_t , and reward r_t describe the decision-making process of the trading agent. Each component of the MDP is presented in detail in the following subsections. Figure 1 shows the architecture of the RL-Based Automated Stock Trading.

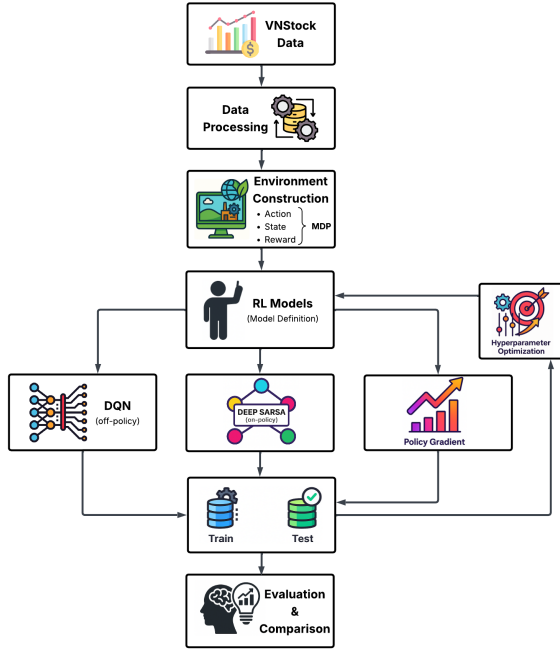


Figure 1. System Architecture for RL-Based Automated Stock Trading

1) *Input Data*: The input data for this study were collected from the VNStock library, an open-source Python library that provides historical trading data for the Vietnamese stock market. Six stocks in the VN30 category [13]: ACB, FPT, GAS, HPG, SSI, and VCB were selected to ensure representativeness, based on three criteria: (i) a continuous listing history since 2013; (ii) inclusion in the VN30 list as of November 2025; and (iii) large market capitalization ($\geq 60,000$ billion VND) and high liquidity ($\geq 1,000,000$ shares/day). The dataset includes OHLCV columns (open, high, low, close, volume) and is normalized by dividing price values by 1,000 VND to improve training stability. During data preprocessing, if any null value is detected, the forward-fill method is applied to maintain the continuity of the time series.

Based on the annual return of the VN-Index [14] during 2013–2025, the study divides the dataset into two market phases: a good phase (2013–2021) with a stable upward trend, and a bad phase (from 2022 onward) when the VN-Index dropped sharply by -32.78% due to post COVID-19 impacts and the corporate bond crisis. Accordingly, two training–testing scenarios are constructed as follows in Table I

To visually illustrate the price dynamics of the selected stocks across the two market phases, Figure 2 below presents

Table I
TRAIN–TEST SPLIT BY MARKET PHASE

Scenario	Train	Test
Before 2022	01/01/2013–31/12/2018	01/01/2019–31/12/2021
After 2022	01/01/2013–31/12/2021	01/01/2022–31/12/2023

the normalized price movements of the six stocks from 2013 to 2024.

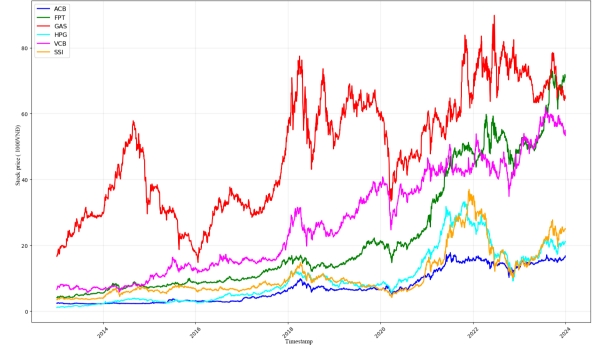


Figure 2. Price movements of six stocks (ACB, FPT, GAS, HPG, VCB, SSI) during 2013–2024

2) *Action*: The action space of the agent in this study consists of buying, holding, or selling the stock at each time step. In this setting, the agent can trade up to k shares per transaction, buying or selling, where $k \in \mathbb{Z}_{>0}$. The action set is defined as follows:

$$A = \{-k, -k+1, \dots, -1, 0, 1, \dots, k-1, k\}.$$

Where:

- k represents the action of buying k shares,
- $-k$ represents the action of selling k shares,
- 0 represents holding (no action) the current position.

This action set allows the model to execute multiple levels of buying and selling, reflecting a more flexible trading strategy rather than being limited to only three simple choices (buy/sell/hold).

3) *State*: In this study, the stock-trading task is modeled as a Markov Decision Process (MDP). The state s is defined at time t as follows:

$$s_t = [p_t, b_t, h_t].$$

Where:

- p_t is the current price of the stock,
- b_t is the current portfolio balance,
- h_t is the number of shares currently held.

To enhance the ability to recognize trends, momentum, and market strength, four technical indicators are integrated into the state space to increase the effectiveness of hand-crafted features in RL trading environments:

$$s_t = [p_t, b_t, h_t, \text{MACD}, \text{RSI}, \text{CCI}, \text{ADX}].$$

4) *Reward*: The reward function is designed to optimize the portfolio value. Specifically, the reward R for the transition from time t to $t + 1$ given action a_t is defined as follows:

$$R(s_t, a_t, s_{t+1}) = (b_{t+1} + p_{t+1} \cdot h_{t+1}) - (b_t + p_t \cdot h_t) - c_t.$$

Where:

- c_t : transaction cost,
- The reward reflects the change in portfolio value after taking action a_t , including both the cash balance and the value of the shares held.

B. Assumptions and Constraints

To simplify the model and ensure the feasibility of the study, several assumptions are made:

- The buy/sell orders are executed at the closing price of the current trading session.
- The RL agent is assumed not to influence the market, meaning it does not affect price or liquidity.
- The transaction fee is assumed to be a small fixed constant applied to each trade:

$$c_t = 0.1\% \cdot p_t \cdot |h_t - h_{t-1}|.$$

- When executing a buy order, the total cost must not reduce the balance below the minimum threshold (e.g., 0, 100 VND).
- When executing a sell order, the number of shares sold cannot exceed the number of shares currently held in the portfolio.

C. Deep Q-Learning Model

The first algorithm used to train the agent is Deep Q-Learning. Since the state space is continuous, $s_t \in \mathbb{R}^7$, and the action set has size $|\mathcal{A}| = 2k + 1$, it is not feasible to construct a traditional Q -table $Q(s, a)$ for all combinations. To address this issue, the study trains a deep neural network to approximate the Q -function.

The Deep Q-Learning training process follows the Algorithm that described in Figure 3 [9]. In each episode, the agent interacts with market data to generate a trajectory, after which the Q -network is updated through backpropagation and optimized using Adam. Huber loss is employed to enhance stability in highly noisy environments. During trajectory generation, the agent selects actions using the ϵ -greedy strategy to maintain a balance between exploration and exploitation. As ϵ gradually decreases over time, the model increasingly prioritizes actions with higher Q -values while still preserving the necessary level of exploration in a non-stationary market.

The neural network used to approximate the Q -function consists of three MLP layers with ReLU activation functions, as illustrated in Figure 4. The input layer includes seven state features (current price, portfolio balance, number of shares held, MACD, RSI, CCI, ADX). The output layer maps to the corresponding Q -values for the entire action space $[-k, \dots, 0, \dots, k]$. The network weights are updated multiple times within each episode to ensure convergence.

Algorithm 1 Deep Q-Learning, Input: α, γ, π, q

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1: Initialize  $\pi$  as uniformly distributed over actions for all states
2: for episode = 1, 2, ... do
3:   Generate a trajectory from current policy  $\pi$ 
4:    $\hat{q}(s, a) = \text{COPY}(q(s, a))$ 
5:   for epoch = 1, 2, ... do
6:     for Each  $s_t, a_t, R_t$  do
7:        $q_{\text{new}}(s_t, a_t) = q(s_t, a_t) + \alpha(R_t + \gamma \max_{a'} \hat{q}(s_{t+1}, a') - q(s_t, a_t))$ 
8:       Loss = HuberLoss( $q(s_t, a_t), q_{\text{new}}(s_t, a_t)$ )
9:       Conduct Backpropagation from Loss
10:      Update weights of  $q$  with Adam Optimizer
11:    end for
12:  end for
13:  Update  $\pi$  with  $\epsilon$ -greedy strategy
14: end for

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Figure 3. Algorithm 1: Deep Q-learning

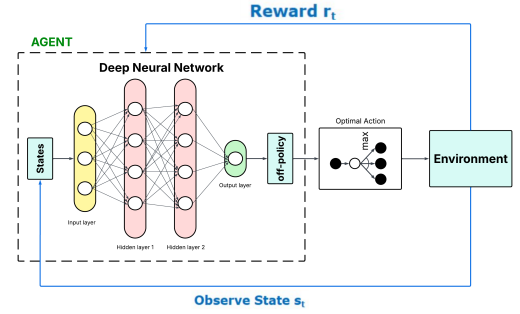


Figure 4. Network Architecture of the Deep Q-Learning Algorithm

D. Deep SARSA

This study conducts experiments with the Deep SARSA algorithm. This model uses the same neural network architecture as Deep Q-Learning but differs in the value update mechanism during training as shown in Figure 5. Instead of relying on the optimal action in the next state, Deep SARSA uses the actual action selected by the agent in the trajectory. As a result, the model more closely reflects the agent's behavior in a non-stationary environment, which is a characteristic of the Vietnamese stock market. In our implementation, the Deep SARSA training process is structured similarly to DQL: the agent interacts with VNStock data to generate trajectories, and the Q-Learning is updated multiple times per episode using Adam and Huber loss. The key difference lies in the update step, where the model uses the pair (s_{t+1}, a_{t+1}) taken directly from the trajectory instead of an assumed optimal action. Hyperparameters such as learning rate, gamma, batch size, and the decay rate of ϵ are fine-tuned through multiple experiments to achieve the best performance. The ϵ -greedy strategy continues to be used to balance exploration and exploitation throughout the training process.

E. Policy Gradient Method

To provide a more intuitive comparison, the study additionally implements the Policy Gradient (PG) method as a baseline model. Unlike DQL and SARSA, PG does not learn the Q -function; instead, it directly learns the parameters of the policy as shown in Figure 6. The neural network outputs a probability distribution over actions, normalized

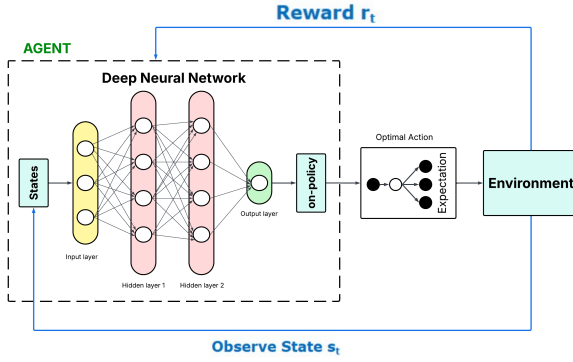


Figure 5. Network Architecture of the Deep SARSA Algorithm

using Softmax. The policy training procedure of the gradient-based model is presented in Figure 7.

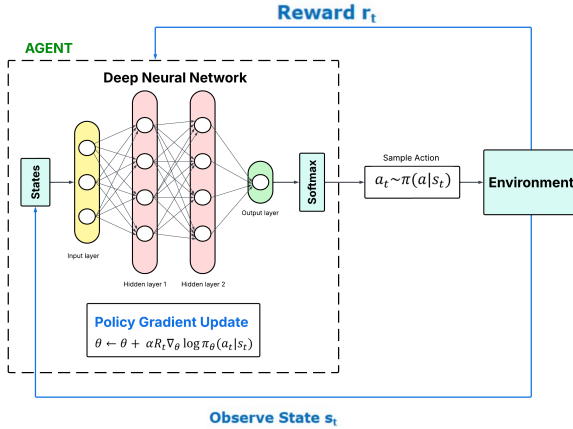


Figure 6. Network Architecture of the Policy Gradient Method

Algorithm 2 Policy Gradient, Input: α, γ, π
Initialize π as uniformly distributed over actions for all states
2: **for** $episode = 1, 2, \dots$ **do**
Generate a trajectory from current policy π
4: **for** $epoch = 1, 2, \dots$ **do**
for Each step t in the episode **do**
6: $G_t = \sum_{k=t}^T \gamma^{k-t} R_k$
end for
8: Loss = $-\frac{1}{2} \sum_t \ln(G_t \pi(a_t | s_t, \theta))$
Conduct Backpropagation from Loss
10: Update weights of π with SGD Optimizer
end for
12: **end for**

Figure 7. Algorithm 2: Policy Gradient

Some notable points:

- The loss has a negative sign because this is a maximization process rather than a minimization one.
- Multiple hyperparameter sets are tested to select the best configuration.
- Finally, the hyperparameters are kept the same across all three models, differing only in the optimization algorithm (Stochastic Gradient Descent (SGD) for PG).

Table II
TRAINING HYPERPARAMETER SETTINGS

Hyperparameter	Default Value	Tuned Value
Episodes	30	35
Discount factor (γ)	0.6	0.95
Learning rate (α)	0.7	0.6
Initial ε	0.8	0.9
Minimum ε	0.2	0.1
ε Decay	0.9	0.95
Epochs for Updating NN	10	10
Learning rate of NN	1×10^{-5}	1×10^{-5}

IV. EVALUATION

A. Evaluation Schema

To objectively and consistently evaluate the performance of the models, the study establishes an evaluation scheme that incorporates market regimes, training—testing configurations, and a comprehensive set of performance metrics.

a) *Definition of Market Regimes:* The study evaluates the models under two distinct market regimes to analyze their behavior under different economic conditions:

- **Favorable period (2013–2021):** A phase of stable and sustained market growth.
- **Adverse period (2022–2023):** A phase of severe market downturn, with the VN-Index experiencing a peak decline of 32.78% in 2022.

b) *Training and Testing Setup:* All algorithms implemented in this study are executed within a unified MDP environment, with the neural network architecture, input features, and reward function kept fixed. Each model is trained and tested independently on each stock and each market regime. The simulated trading environment supports three actions (BUY, SELL, HOLD), applies a transaction fee of 0.1%, and computes rewards based on changes in portfolio value. The hyperparameters are presented in Table II, and each model is run 20 times to assess the stability and reliability of the results.

c) *Performance Evaluation Metrics:* The performance of the models is evaluated based on two key profitability metrics:

- **Profit:** Represents the absolute profit earned after the trading cycle.

$$\text{Profit} = V_{\text{final}} - V_{\text{initial}}.$$

- **Annualized Rate of Return (ARR):** Enables consistent comparison of growth rates across models regardless of differences in market regime duration. Here, N denotes the number of years in the trading period.

$$\text{ARR} = \left[\left(\frac{V_{\text{final}}}{V_{\text{initial}}} \right)^{\frac{1}{N}} - 1 \right] \times 100.$$

B. Before 2022: The Period of Development

To evaluate the effectiveness of the algorithms during the favorable market phase, the study analyzes trading outcomes

across six stocks. Table III below presents the profitability of the three models for each stock and for the overall portfolio.

The analysis of Table III during the favorable market phase (before 2022) shows a clear superiority of the Deep SARSA algorithm compared with the other two methods. Specifically, Deep SARSA achieves a total profit of 6,042.61 million VND and an annualized rate of return (ARR) of 25.05%, which is noticeably higher than Deep Q-Learning (4,908.00 million VND, ARR 20.97%) and Policy Gradient (4,812.58 million VND, ARR 20.66%). A notable highlight is that Deep SARSA not only leads overall but also demonstrates exceptional stability, generating positive returns across all six stocks, including highly volatile ones such as GAS and HPG. These results reflect the model's ability to learn optimal policies and effectively adapt to a prolonged upward market trend. In contrast, Deep Q-Learning exhibits inconsistent performance: it performs well on certain stocks such as FPT, HPG, and SSI, but underperforms on others. This decline in overall performance may stem from the off-policy nature of the algorithm, which is prone to overly optimistic Q -value estimates and reduced stability in highly continuous environments such as a long-term bull market. Regarding Policy Gradient, although it achieves profits comparable to Deep Q-Learning on some stocks, it ultimately yields the lowest overall performance. This suggests that the model is more sensitive to noise and faces challenges in maintaining a consistent action policy, potentially due to the high-variance nature of direct policy learning (policy-based methods).

Overall, the experimental results reinforce the argument for the advantage of on-policy mechanisms under stable market conditions. Deep SARSA, with its value-update mechanism based on the agent's actual actions, demonstrates not only superior profit maximization but also significantly higher stability and reliability compared with off-policy approaches (Deep Q-Learning) or direct policy-based methods (Policy Gradient) within the scope of this study.

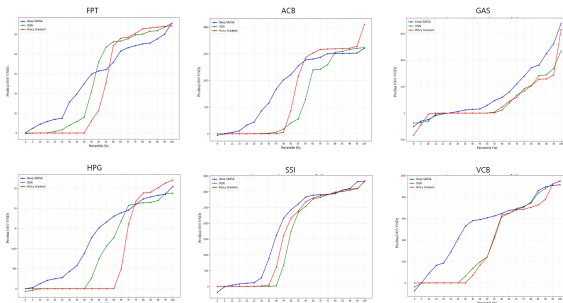


Figure 8. Profit Distribution of the Models for Each Stock over 20 Runs

The profit distribution charts in Figure 8 show that Deep SARSA maintains the most stable performance across all 20 runs for every stock. Its distribution curves are consistently shifted to the right and rise steadily even from the lower percentiles, indicating strong reproducibility of returns and low sensitivity to noise. In contrast, Deep Q-Learning ex-

hibits a wider distribution, reflecting the inherent variability of its off-policy mechanism, while Policy Gradient records numerous runs with very low profit, especially in the 0–40% percentiles, highlighting its high level of instability. Overall, the distribution patterns indicate that SARSA is both stable and effective, whereas DQL and PG display less consistent performance.



Figure 9. Portfolio Value Over Time Chart

The portfolio value trajectories in Figure 9 show that Deep SARSA consistently achieves the highest and smoothest capital growth across most stocks, with particularly strong performance on FPT, HPG, and SSI. SARSA closely follows the market's upward trend with minimal sudden drawdowns, reflecting the advantage of its on-policy update mechanism in an uptrending environment. Deep Q-Learning ranks second, producing reasonably strong capital curves but with more pronounced fluctuations, indicating reduced stability during short-term corrections. Policy Gradient typically yields the lowest capital levels with high volatility, especially for GAS and HPG, suggesting difficulty in maintaining a consistent trading strategy.

C. From 01/01/2022 to 31/12/2023: The Period of Decline

To evaluate the adaptability of the models under declining market conditions, we assess the performance of the three algorithms during the "Bad Situation" period of 2022–2023.

The Table IV presents the Profit and ARR results of each model across the six stocks in a downtrend market context. The values indicate that all three algorithms experience a significant decline in performance during the 2022–2023 recession period, reflecting the strong impact of a prolonged downturn and unpredictable volatility. Deep SARSA remains the top-performing model, achieving a total profit of 236.06 million VND and an ARR of 1.77%, despite incurring losses on three stocks (GAS, HPG, and SSI). This suggests that SARSA still maintains relatively effective performance under adverse market conditions, primarily due to its ability to capitalize on recovery phases in FPT and VCB. Deep Q-Learning yields a lower total profit (187.57 million VND, ARR 1.44%), although its losses on sharply declining stocks (such as GAS, SSI, and HPG) are smaller than those of SARSA, reflecting a more "defensive" tendency of the off-policy mechanism when the market reverses. Policy Gradient

Table III
RETURNED PROFITS ON THE TEST SET BEFORE 2021, VALUES IN MILLION VND

Method	FPT	ACB	GAS	HPG	SSI	VCB	Total
Deep SARSA Profit (Annual Rate)	1,404.28 (34.03%)	906.28 (24.04%)	150.14 (4.78%)	1,271.82 (31.52%)	1,784.09 (40.76%)	526.00 (15.16%)	6,042.61 (25.05%)
Deep Q-Learning Profit (Annual Rate)	1,364.00 (33.29%)	611.11 (17.26%)	84.66 (2.75%)	973.63 (25.48%)	1,492.54 (35.65%)	381.77 (11.40%)	4,908.00 (20.97%)
Policy Gradient Profit (Annual Rate)	1,194.26 (30.00%)	767.39 (20.94%)	86.89 (2.82%)	799.92 (21.68%)	1,599.93 (37.58%)	364.19 (10.93%)	4,812.58 (20.66%)

Table IV
RETURNED PROFITS ON THE TEST SET FROM 2022 TO 2023, VALUES IN MILLION VND

Method	FPT	ACB	GAS	HPG	SSI	VCB	Total
Deep SARSA Profit (Annual Rate)	294.60 (13.91%)	49.80 (2.48%)	-26.63 (-1.35%)	-89.27 (-4.61%)	-80.07 (-4.12%)	87.63 (4.33%)	236.06 (1.77%)
Deep Q-Learning Profit (Annual Rate)	239.89 (11.46%)	33.10 (1.66%)	-32.82 (-1.67%)	-65.27 (-3.35%)	-56.72 (-2.90%)	69.39 (3.44%)	187.57 (1.44%)
Policy Gradient Profit (Annual Rate)	229.54 (10.99%)	42.02 (2.10%)	2.34 (0.12%)	-72.43 (-3.72%)	-53.84 (-2.75%)	74.82 (3.71%)	222.45 (1.74%)

achieves moderate performance (222.45 million VND, ARR 1.74%), showing a notable advantage on GAS by maintaining positive returns, but it still exhibits instability on stocks with strong downward trends.

Overall, the results show that the adaptability of RL models deteriorates significantly when the market transitions into a downtrend, with performance gaps between the three algorithms narrowing compared with the favorable period. Deep SARSA remains the top performer in terms of overall profit, whereas Deep Q-Learning demonstrates better risk control on sharply declining stocks, and Policy Gradient exhibits intermediate behavior but lacks stability. This clearly reflects the impact of market regime shifts on the policy-update mechanisms of each algorithm.

tial differences compared with the favorable market phase. Deep SARSA continues to achieve high returns at the upper percentiles but exhibits greater volatility, particularly for sharply declining stocks such as HPG, GAS, and SSI. Deep Q-Learning shows a narrower distribution across several stocks, reflecting its tendency to adopt defensive (hold-biased) strategies in adverse market conditions; however, this behavior also causes the model to miss many recovery opportunities. Policy Gradient displays a volatile distribution but occasionally outperforms at the higher percentiles especially for GAS though overall its results remain inconsistent. These findings indicate that in a downtrend environment, none of the models maintain strong consistency, and the level of risk increases noticeably across all three algorithms.

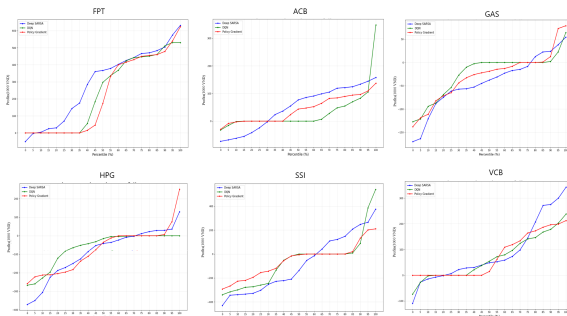


Figure 10. Profit Distribution of the Models for Each Stock over 20 Runs

The profit distribution in Figure 10 charts reveal substan-



Figure 11. Portfolio Value Over Time Chart

The portfolio value curves in Figure 11 illustrate the

strong impact of the market downturn on all three models. For stocks such as FPT and VCB, Deep SARSA maintains relatively strong capital trajectories and can capture recovery phases, but it exhibits deeper declines for highly volatile stocks like HPG and GAS. Deep Q-Learning generally produces smoother capital curves, as its off-policy mechanism helps limit losses during sharp downturns, a pattern particularly evident in HPG and SSI. Policy Gradient shows considerable fluctuations and struggles to maintain a stable trajectory, although in certain periods (such as for GAS) it demonstrates a better ability to mitigate losses.

V. CONCLUSION

This study evaluates the effectiveness of three reinforcement learning algorithms: Deep SARSA, Deep Q-Learning, and Policy Gradient in the context of stock trading in the Vietnamese market, across two contrasting market regimes: a stable growth period (2013—2021) and a severe downturn (2022—2023). The experimental results show that Deep SARSA consistently achieves the highest performance, with superior profitability and greater stability in favorable market conditions. In contrast, Deep Q-Learning demonstrates better risk control during the recession period, while Policy Gradient records the lowest and most unstable performance across both market regimes. This discrepancy clearly reflects the impact of market regimes on the update mechanisms of each algorithm: on-policy models like SARSA adapt more effectively when the market follows a clear trend but react more slowly to sudden reversals, whereas off-policy methods such as DQL tend to be more defensive yet struggle to capitalize on strong upward movements. At the same time, the results highlight a common limitation across all three algorithms when facing regime shifts, particularly in prolonged downtrend markets, indicating that learning from historical data alone does not guarantee effectiveness when the state distribution changes significantly. Future work will focus on expanding the dataset to include additional market cycles, integrating macroeconomic factors, and developing adaptive RL approaches to improve responsiveness under highly volatile market conditions.

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